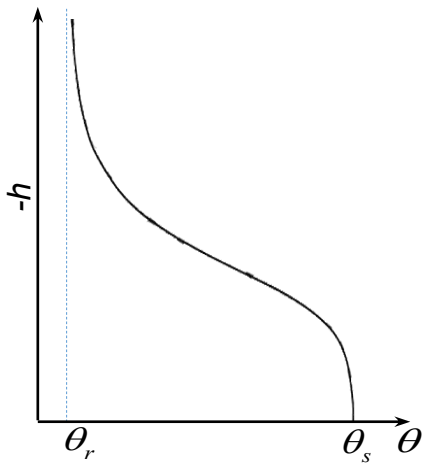
HWRS 561b: Physical Hydrogeology II

PFAS (part 2)

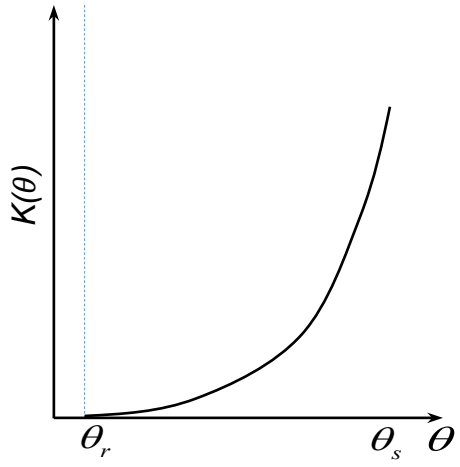
Modeling PFAS Transport in the Vadose Zone

Variably saturated flow in the vadose zone

$$\frac{\partial \theta}{\partial t} - \nabla \cdot [\boldsymbol{K}(\theta)\nabla(h - z)] = 0$$



SWC



Unsat hydraulic conductivity

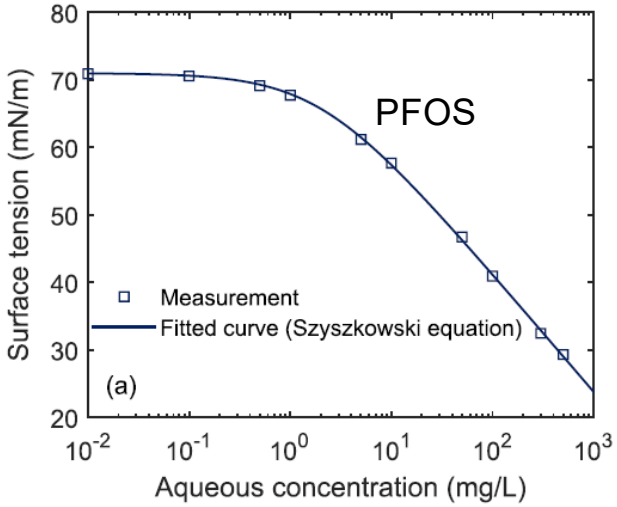
PFAS dissolving in water changes surface tension

$$\sigma = \sigma_0[1 - b\ln(1 + \frac{C}{a})]$$

PFAS concentration

Surface tension of water w/o PFAS

Fitting parameters



Question: Will the SWC and $K - \theta$ change in the presence of PFAS? If so, how?

Modeling PFAS Transport in the Vadose Zone

Let's employ the capillary tube model to think about the problem (Leverett-J scaling)

w/o PFAS

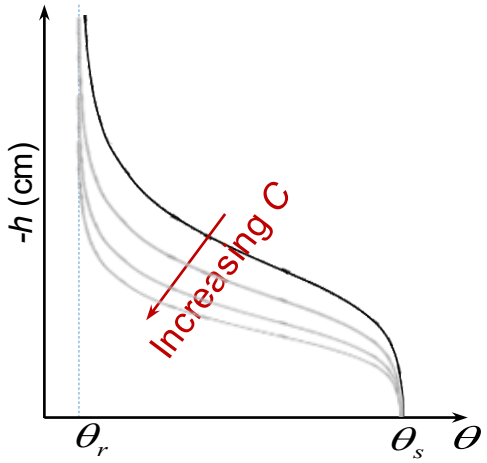
$$-h_0 = \frac{2\sigma_0 \cos\gamma_0}{\rho g r}$$

w/ PFAS

$$-h = \frac{2\sigma \cos\gamma}{\rho g r}$$

$$\Rightarrow h = \frac{\sigma \cos\gamma}{\sigma_0 \cos\gamma_0} h_0 \quad \Rightarrow \theta = \begin{cases} \theta_r + \dfrac{\theta_s - \theta_r}{\left[1 + \left(\dfrac{\sigma_0 \cos\gamma_0}{\sigma \cos\gamma} \alpha |h|\right)^n\right]^m}, h < 0 \\ \theta_s, h \geq 0 \end{cases}$$

Scaling factor



Insights:

(1) We can think of this as PFAS modifies the α in the V-G model.

$$\alpha_{\text{modified}} = \frac{\sigma_0 \cos\gamma_0}{\sigma \cos\gamma} \alpha$$

(2) Now, the variably saturated flow depends on PFAS transport.

Modeling PFAS Transport in the Vadose Zone

PFAS transport in variably unsaturated flow (neglecting the interactions among PFAS)

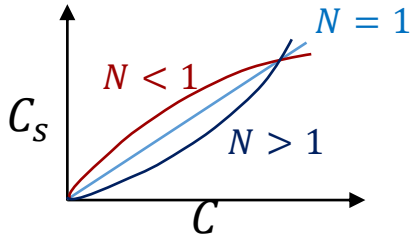
Advection Dispersion

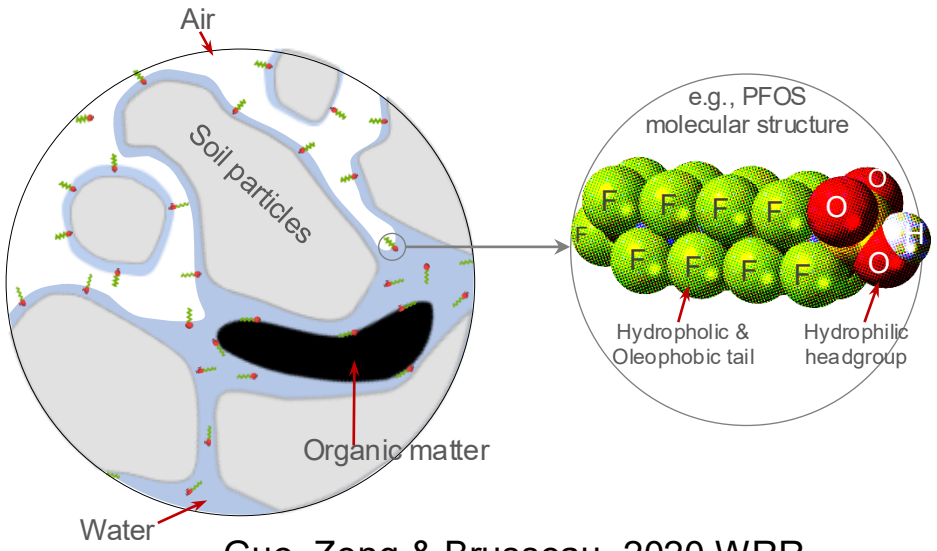
$$\frac{\partial(\theta C)}{\partial t} + \rho_b \frac{\partial C_s}{\partial t} + \frac{\partial C_{aw}}{\partial t} + \nabla \cdot (\theta \boldsymbol{v} C) - \nabla \cdot (\theta \boldsymbol{D} \nabla C) = 0$$

Storage in the aqueous phase

Storage in the solid-phase adsorption

Storage in the air-water interfacial adsorption





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Solid-phase adsorption $C_s = K_f C^N$

Air-water interfacial area

Air-water interfacial adsorption $C_{aw} = \Gamma A_{aw} = K_{aw} C A_{aw}$ $\sigma = \sigma(C)$ Szyszkowski equation

$$K_{aw} = \frac{\Gamma}{C} = -\frac{1}{RTC}\left(\frac{\partial \sigma}{\partial \ln C}\right)_T$$

$$\Rightarrow K_{aw} = \frac{1}{RT}\frac{\sigma_0 b}{a + C}$$

Modeling PFAS Transport in the Vadose Zone

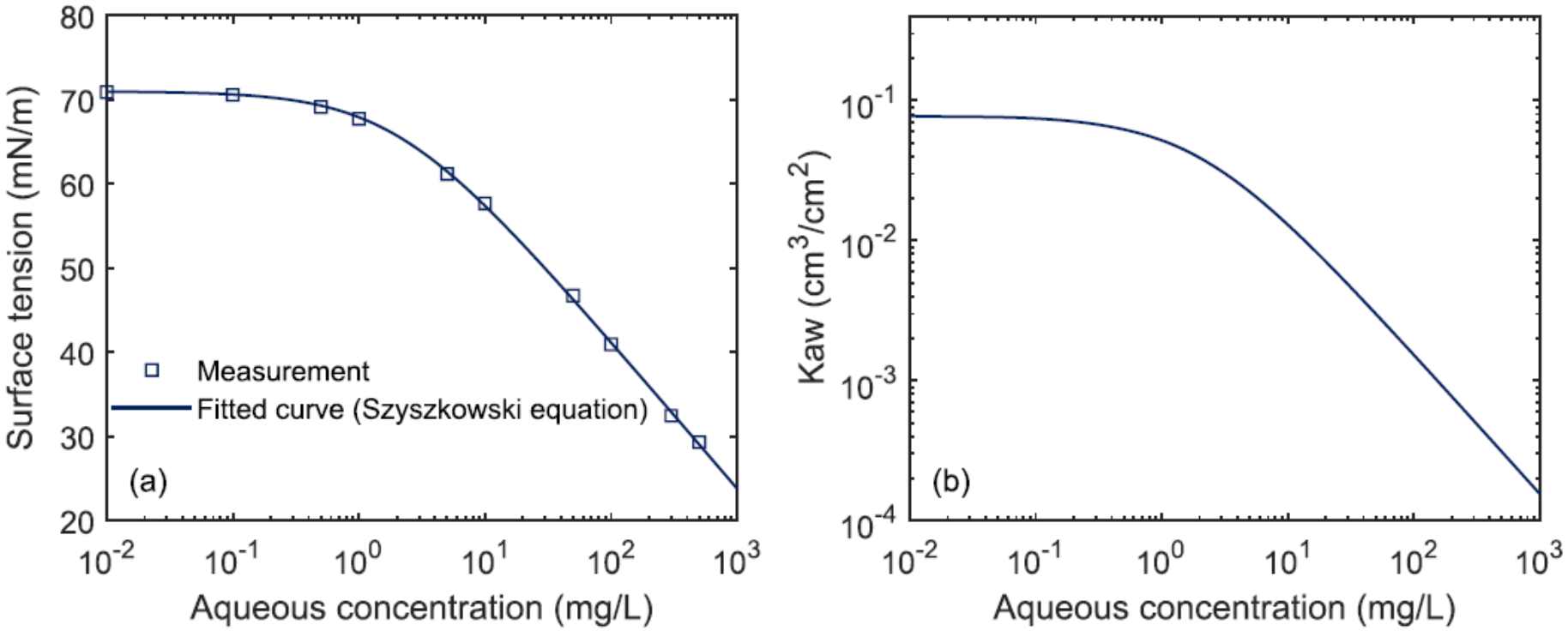

Figure 3. (a) Measured surface tension as a function of the aqueous concentration of PFOS. The solid line is a fitted curve using the Szyszkowski equation. (b) The computed air-water interfacial adsorption coefficient (K_{aw}) as a function of the aqueous concentration of PFOS.

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Simplified Models

- Steady-state unsaturated flow:

$$q = K$$

- PFAS transport:

$$\frac{\partial(\theta C)}{\partial t} + \rho_b \frac{\partial C_s}{\partial t} + \frac{\partial C_{aw}}{\partial t} + \nabla \cdot (\theta \boldsymbol{v} C) - \nabla \cdot (\theta \boldsymbol{D} \nabla C) = 0$$

Advection Dispersion

Linear and rate-limited solid-phase adsorption

$$C_{s,1} = F_s K_d C$$

Instantaneous

$$\frac{dC_{s,2}}{dt} = \alpha_s[(1 - F_s)K_d C - C_{s,2}]$$

Kinetic

Linear and equilibrium air-water interfacial adsorption

$$C_{aw} = A_{aw} K_{aw} C$$

Instantaneous